



CSE422: Artificial Intelligence

Lab Project Report Consumer Classification

Group 5-

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1. Introduction

The project is about the prediction of consumer classification churn. This project aims to predict whether a consumer will churn or not, based on the given behavioral features such as age, income, marital status, credit score, number of purchases, membership years, and device used.

The motivation behind this project is that consumer churn is a major issue in many industries, as it directly affects profitability. Predicting churn enables organizations to take preventive measures, such as offering promotions or enhancing customer engagement.

2. Dataset Description

Number of features:

The dataset contains around 8 original features, which include demographic information, behavioral attributes.

Problem Type

This is a classification problem because the target variable 'Churn' is discrete with two possible values:

- 0 = Not churn
- 1 = Churn

Number of Data Points:

The dataset contains approximately 1500 rows.

Types of Features:

- Quantitative (Numerical): Age, Income, Credit_Score, Num_Purchases, Membership_Years.
- Categorical: Gender, Marital_Status, Device_Used.

Need for Encoding:

Yes. Since machine learning algorithms cannot directly interpret categorical variables, they must be converted into numerical form. One-Hot Encoding was used for this purpose.

Feature Correlation:

A correlated heatmap was applied to all numerical features. The results showed-

- Weak correlations between most input features and the target (Churn). For example, Credit_Score and Income were only slightly related to churn,
- Overall, the dataset lacks strongly predictive features, which partially explains the low model performance.

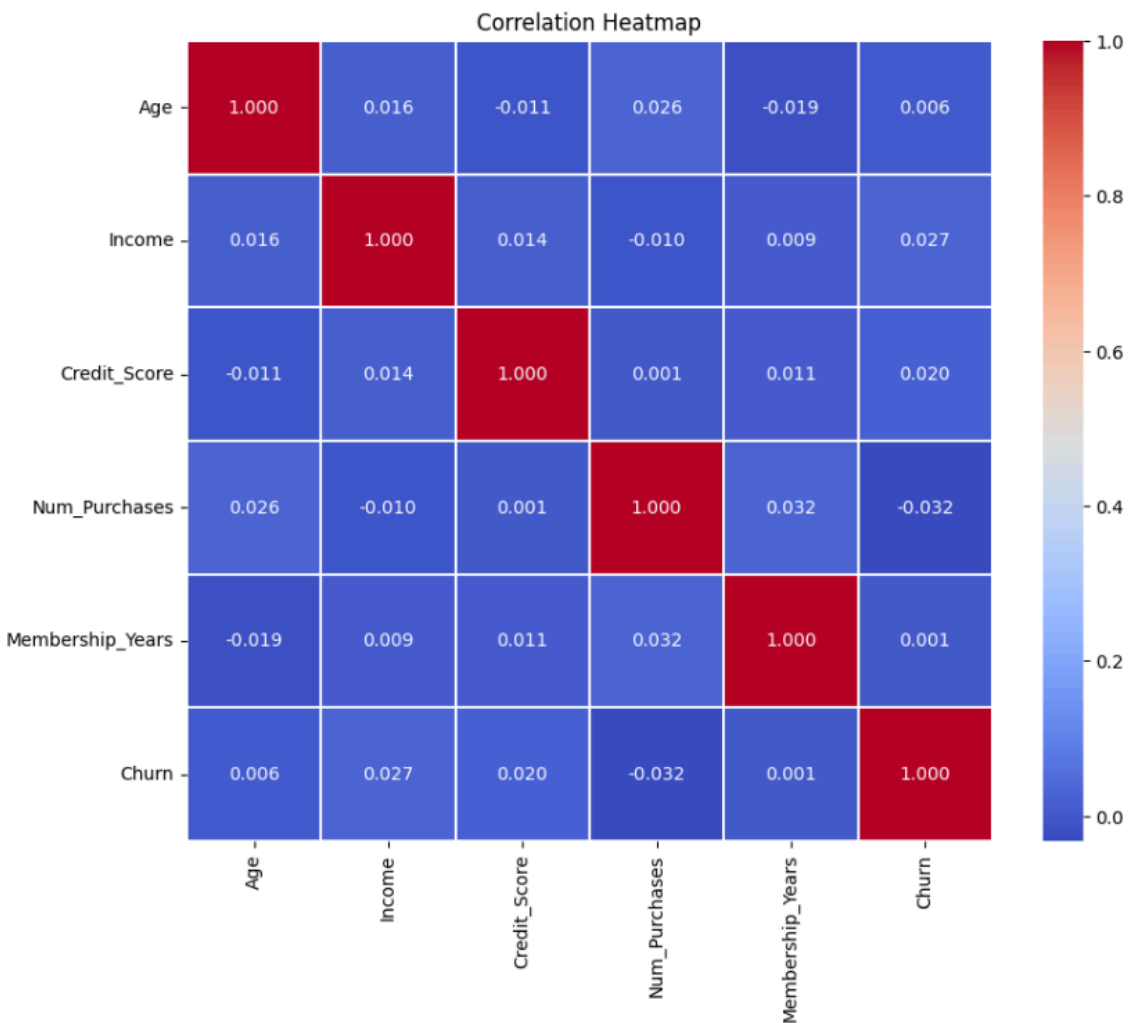
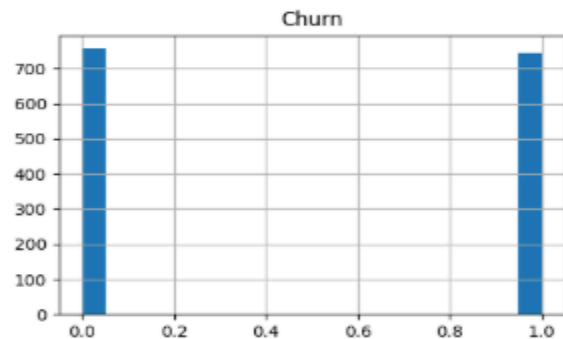
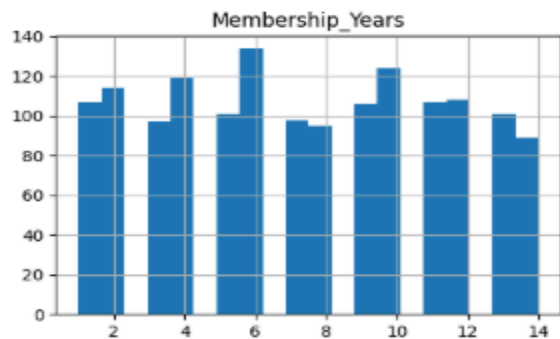
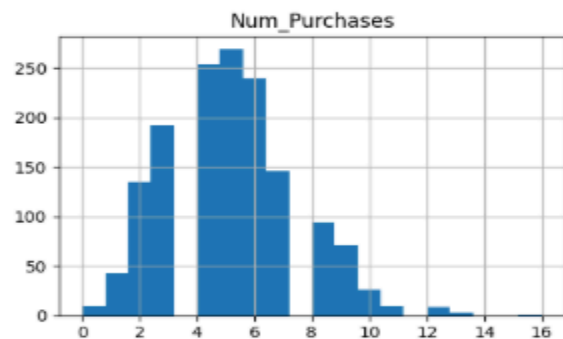
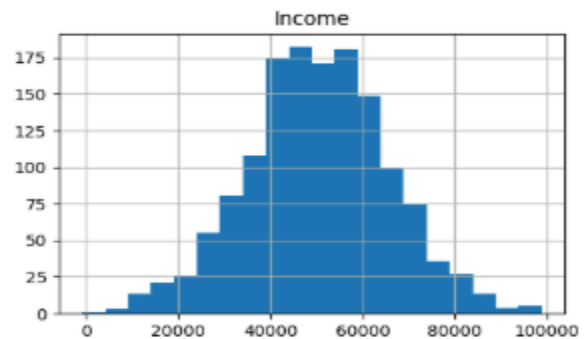
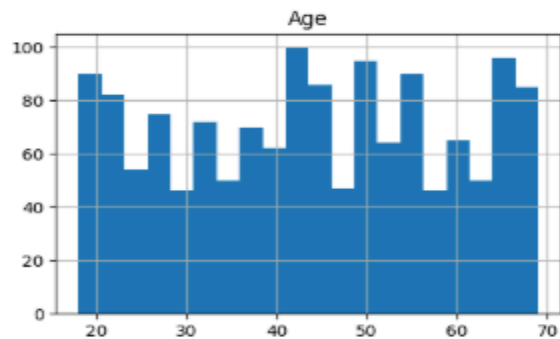
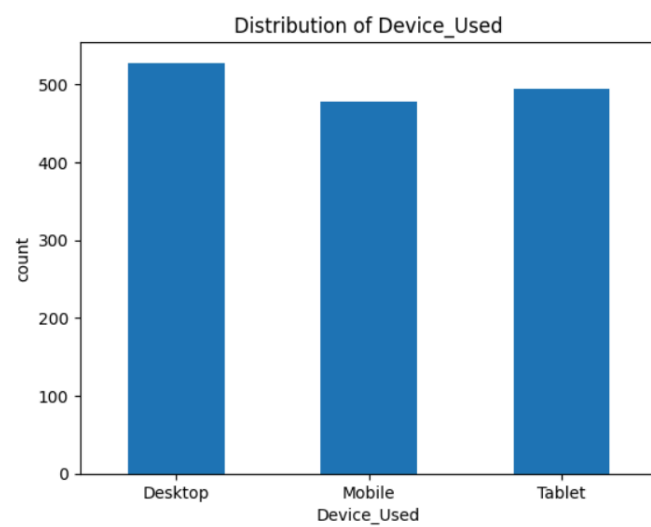
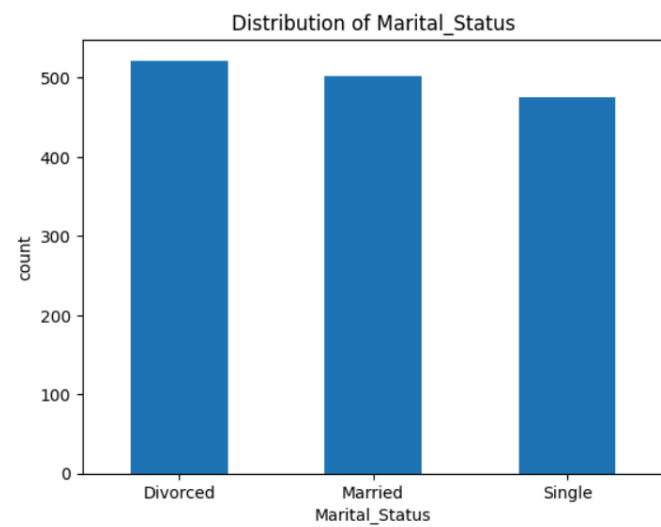
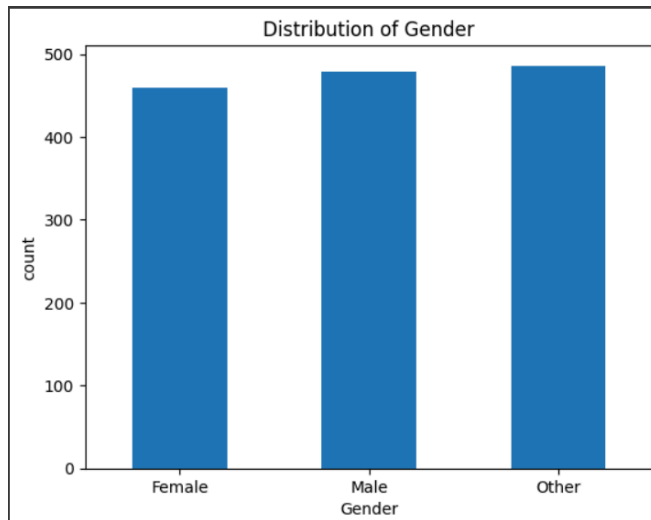


Figure: Correlation Heatmap

Imbalance Dataset Check:

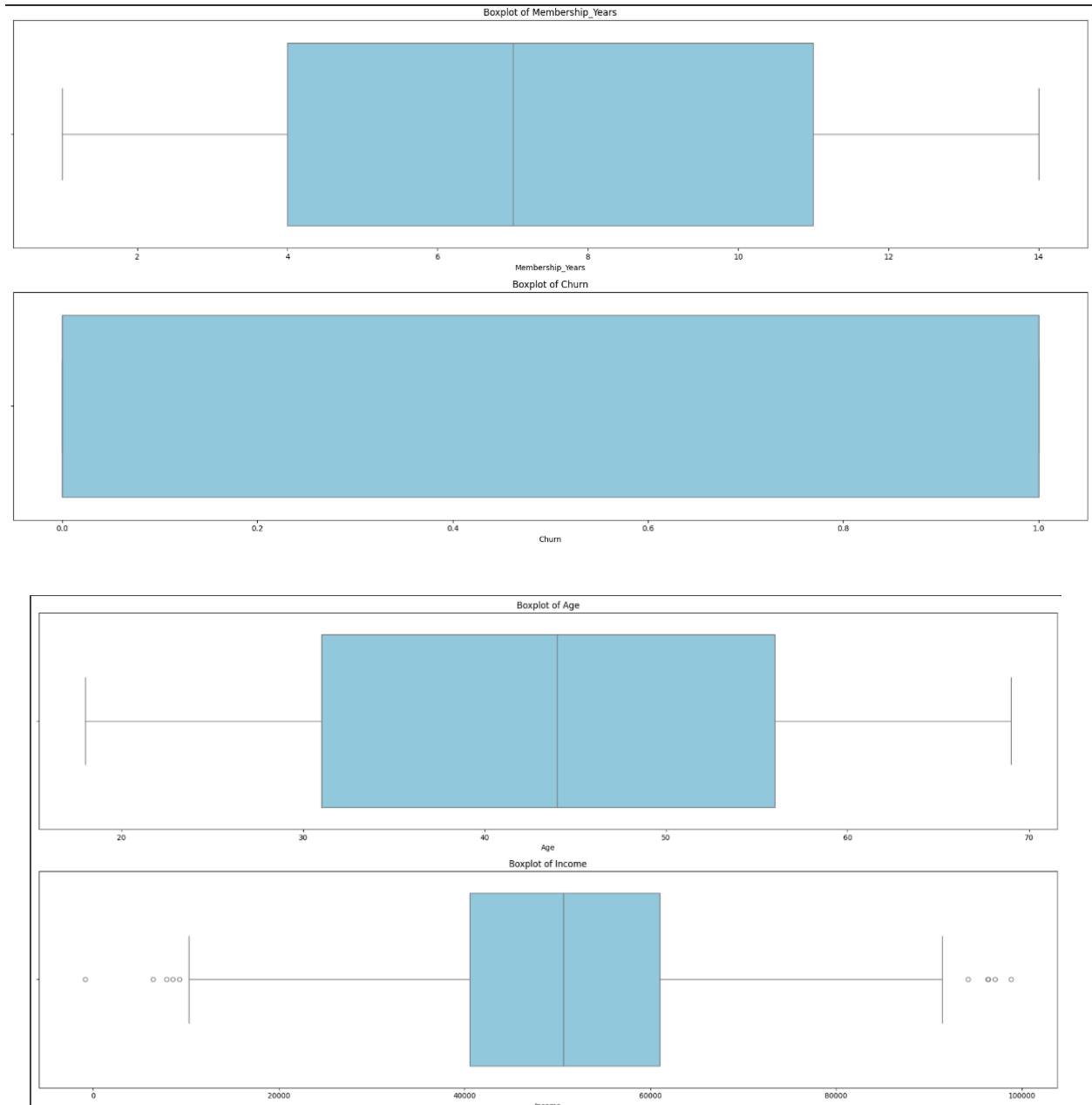
The target variable has two classes: churned (1) and not churned (0). A bar chart confirmed that both classes are nearly equal in size, so the dataset is not significantly imbalanced. The means of oversampling/undersampling techniques were not required.

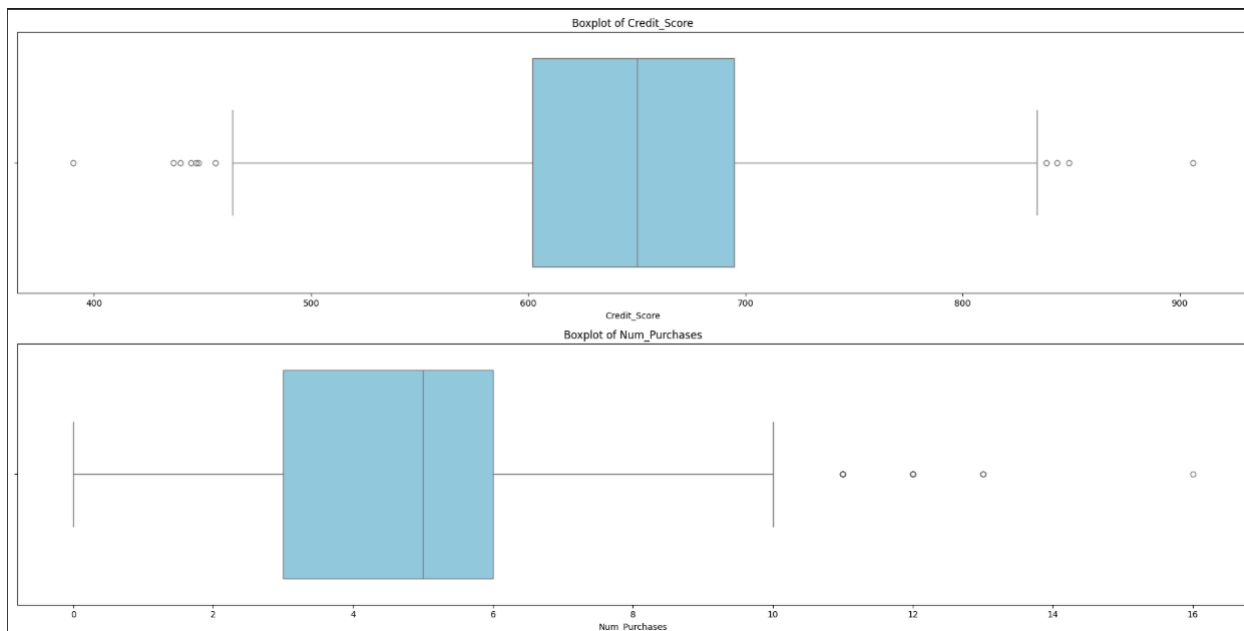




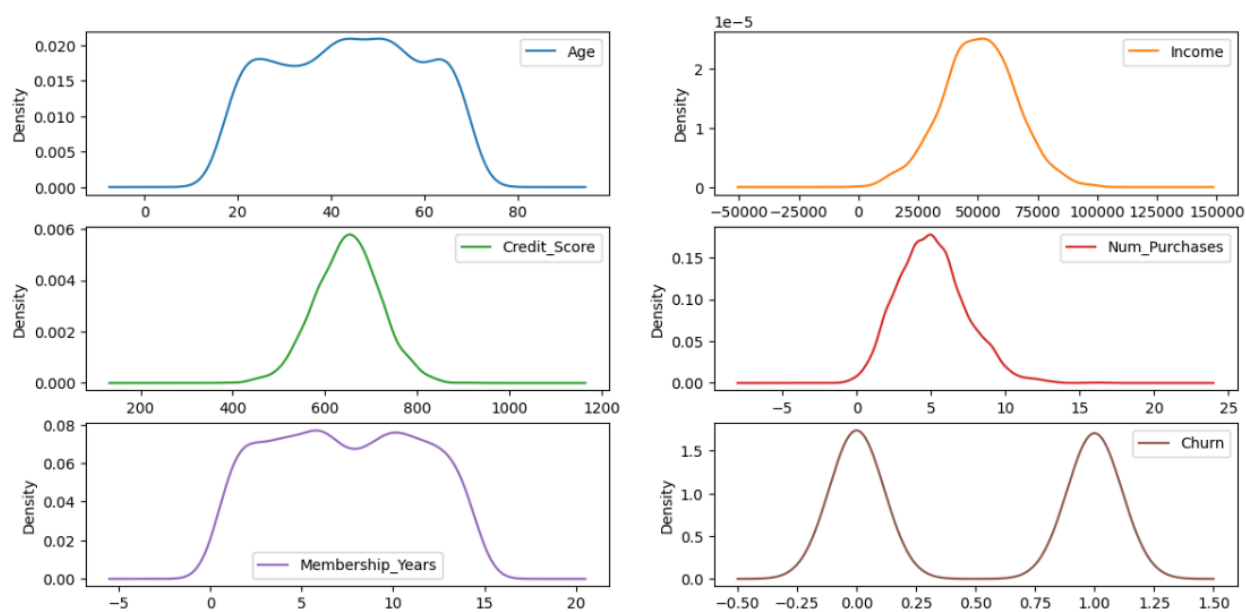
Exploratory Data Analysis(EDA):

Distribution plots showed skewness in Income and Number of Purchases. Boxplots identified outliers in Income and Age. Purchases_per_year and Income_per_purchase highlighted consumer spending behavior, but still did not show a strong separation between churned and non-churned groups. However, EDA confirms that feature overlap makes churn prediction difficult with this dataset.





Density plot of Numerical features



3. Dataset pre-processing

Fault 1: Null/Missing Values

Problem: Some numerical values were invalid, such as ages <0 or >120 , credit scores outside 300-850, and negative membership years. These values were marked as missing values.

Solution: Numerical features were imputed using the KNN imputer, which estimates missing values based on the values of similar records. Categorical features such as gender, Marital_Status, and Device_Used were imputed using the Most Frequent strategy to fill missing entries.

Fault 2: Categorical Values

Problem: The dataset contained categorical features (Gender, Marital_Status, and Device_Used), which machine learning models cannot directly process.

Solution: These features were transformed into the numerical format using One-Hot Encoding, which creates binary indicator variables for each category. This ensured that categorical attributes could be properly used by the models

Fault 3: Feature Scaling

Problem: The numerical features had different ranges (such as Income vs Age vs Credit_Score). Algorithms like KNN and Neural Networks are sensitive to feature scale, so this inconsistency would bias the models.

Solution: All numerical features were scaled using MinMaxScaler, which transforms them to a fixed range (typically 0 to 1). This preserves the shape of the distribution but adjusts the scale, ensuring that all features contribute proportionally and no variable dominates due to larger values.

4. Dataset Splitting

Train: 70%

Validation: 15%

Test: 15%

We split the data into training and testing sets using stratified sampling to preserve the class distribution. About 70% of the data was used for training, 15% for validation, and the rest 15% for testing.

5. Model Training & Testing (Supervised)

We have used a total of three models, the mandatory one is the Neural Network, and the other two are KNN and Decision Tree.

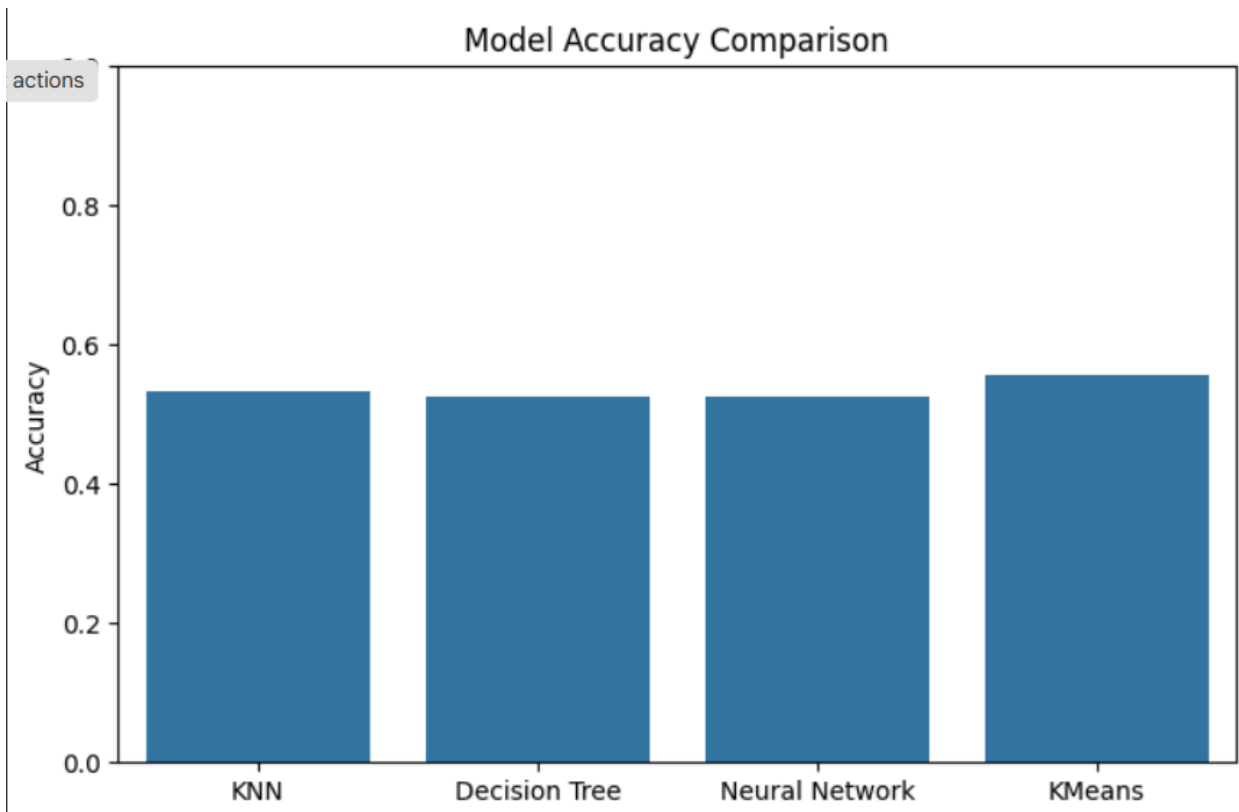
- a. K-Nearest Neighbors (KNN)
 - Tuned $k=5$
 - Accuracy ~ 0.53 (Test Set)
 - Precision/Recall: Balanced but weak (~ 0.50)
 - KNN struggled because classes overlap in the feature space
- b. Decision Tree
 - Tuned $\text{max_depth}=5$, $\text{min_samples_split}=10$
 - Accuracy ~ 0.5344
 - Tended to overfit if the depth was large
 - Could not capture strong predictive patterns
- c. Neural Network (MLPClassifier)
 - Hidden layers: $[(32,), (64,), (64, 32)]$
 - Activation: ReLU
 - Optimizer: Adam, Max iterations: 500
 - Accuracy ~ 0.5244
 - Close to random guessing $\rightarrow$ dataset features not strongly predictive
- d. K-Means(Unsupervised)

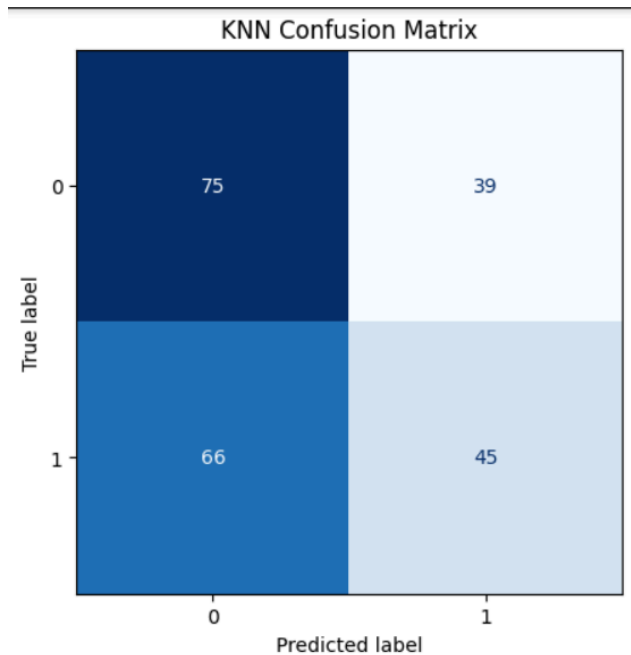
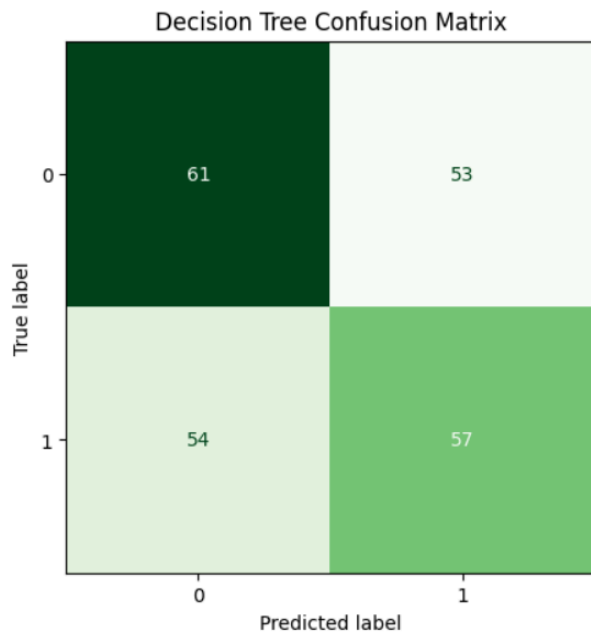
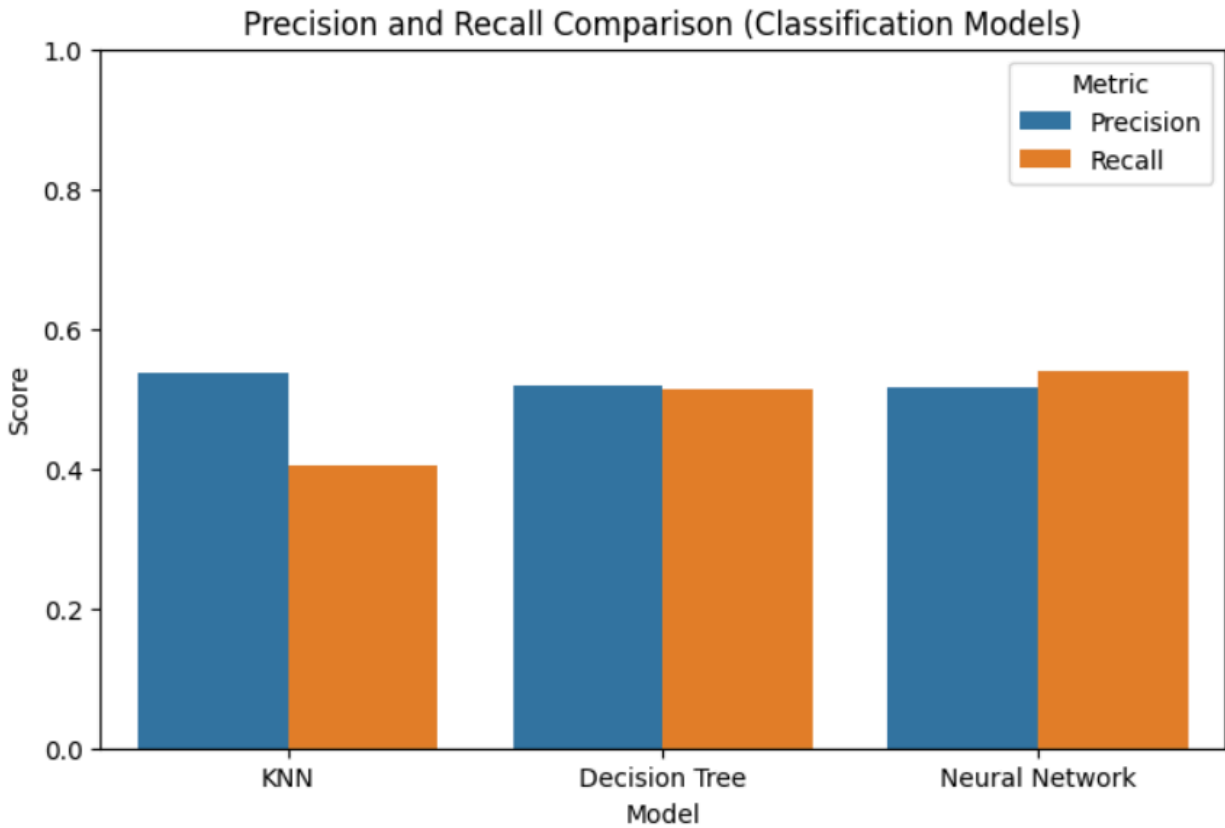
Applied K-Means clustering with the number of clusters selected using the silhouette Score ($k = 2$ was optimal)

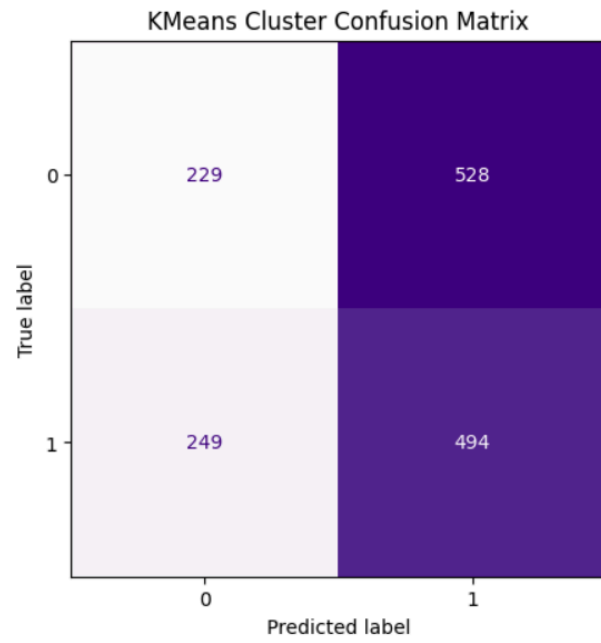
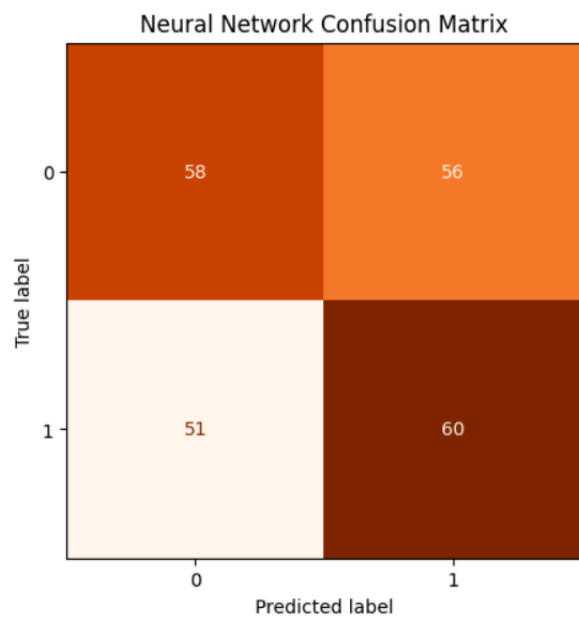
- Silhouette Score: Low(0.3235), indicating poor separation of clusters.
- When the clusters were mapped to churn labels by majority vote (pseudo-accuracy):
 - Validation accuracy ~ 48%
 - Test Accuracy ~ 50%
- Confirms that natural clusters in the data do not align well with the churn labels.

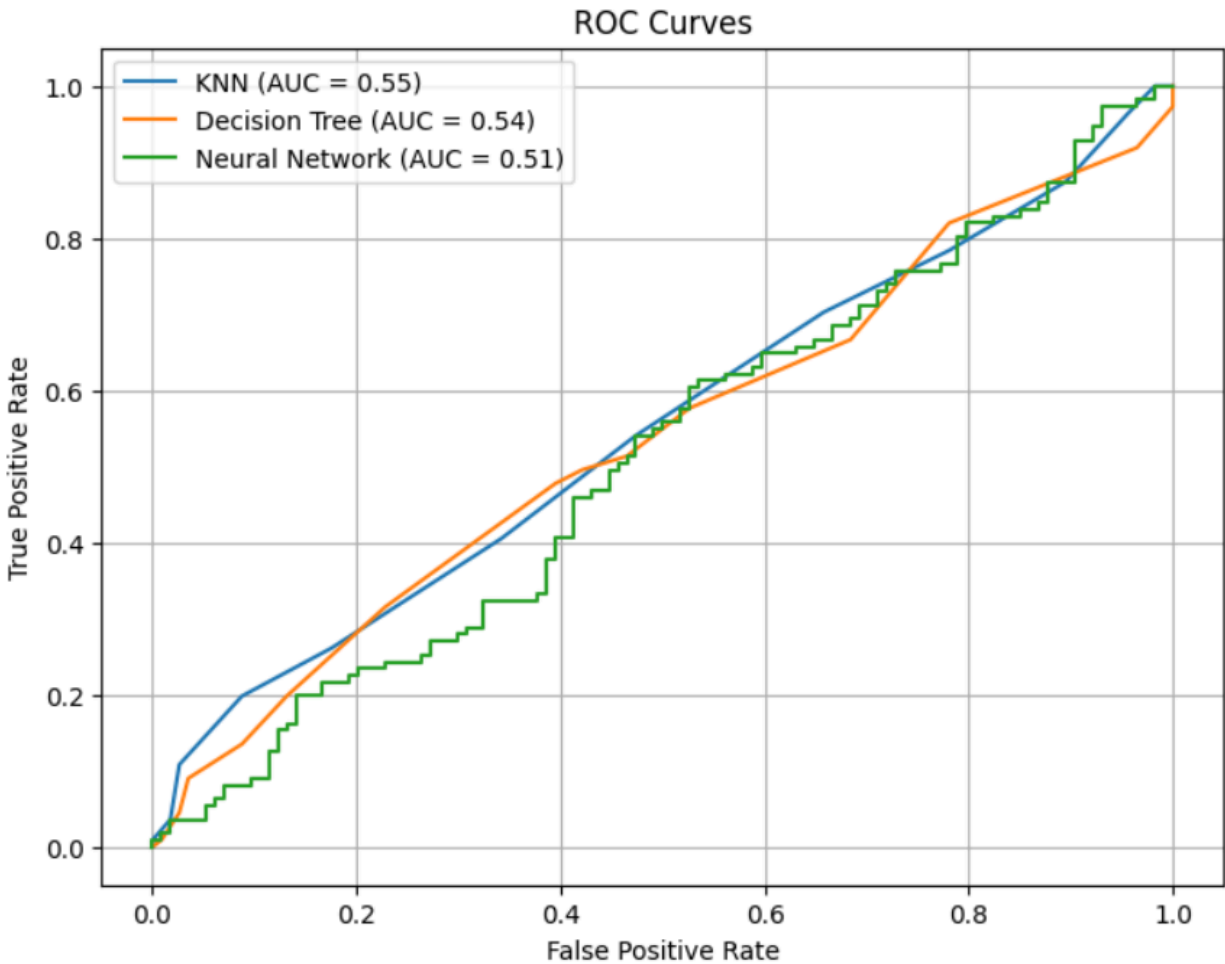
6. Model Comparison & Analysis

- Accuracy Results (Test Set):
 - KNN ~ 53%
 - Decision Tree ~ 53%
 - Neural Network ~ 52%
- All models performed near 50%, indicating weak predictive signals in the dataset. Precision, recall, and F1-scores were consistently ~ 0.50-0.53 across all models. ROC-AUC ~ 0.50, which confirms poor separability.









7. Conclusions

We successfully implemented the full ML pipeline: preprocessing, model training, validation, and testing. The dataset's features were not strong enough predictors of churn, as models hovered around 50% accuracy. Every model was very close to each other. The main challenge was dataset quality, not implementation. Because our accuracy was below 50%. So, we had to make several changes to push the accuracy over 50%. But it could be improved using advanced methods like Random Forest or Gradient Boosting, etc. Regardless, even with modest accuracy, this project reflects what was expected from us, which was the main goal.